

The Cooking Recipe Hub

Final Project Report Interactive Web Application Development

**I Year | II Semester
ICT 1209 - Web Technologies**

**Department of ICT
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Group No: 17

Group Members:

L. P. D. T. Senarathna - ITT/2024/099
R. M. D. S. Kumara - ITT/2024/056

Submitted To:

Ms. Piumi N. Herath
Lecturer (Probationary), Department of ICT
Faculty of Technology, Rajarata University of Sri Lanka

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1. Project Overview

1.1 Purpose of the Project

Cooking at home is something many people enjoy, but managing recipes in daily life can get difficult. People often write down recipes on random pieces of paper, save screenshots on their phones or bookmark different websites. Because of this, it is easy to forget ingredients, lose track of good meals or struggle to decide what to cook for dinner.

To solve this practical issue, we created “The Cooking Recipe Hub” as our mini project for the ICT 1209 (Web Technologies) course. Our main goal is to build an easy-to-use web application where people can find different food ideas, sort recipes by categories and share their own favorite dishes with others.

1.2 Target Audience

Our web application is built for people who want a simple way to explore and organize recipes:

- **Home Cooks:** People who prepare daily family meals and want an organized collection of tasty dishes.
- **Students and Working People:** Busy individuals who need quick ideas for breakfast, lunch or snacks with clear steps.
- **Sri Lankan and Global Food Lovers:** Anyone wanting to prepare classic Sri Lankan food (like Kiribath, Polos Curry, Fish Ambul Thiyal and Konda Kevum) as well as popular international meals.
- **Community Members:** Users who enjoy cooking and want to upload their own recipes along with pictures.

1.3 Key Features

The core features of our website include:

- **User Registration and Login:** New users can create an account with a username, email and password. Registered users can log in securely and log out anytime.
- **Home Page Carousel:** An interactive image slider on the home page displaying traditional Sri Lankan dishes with automatic sliding.
- **Recipe Catalog and Filtering:** A dedicated recipes page where users can browse dishes and filter them by category (All, Breakfast, Brunch, Lunch, Dinner, Snack and Dessert) using JavaScript.
- **Live Recipe Search:** A search bar that filters recipe cards in real time as the user types without refreshing the page.
- **User Dashboard for Recipe Uploads:** A private dashboard where logged-in users can fill a form to submit new recipes along with an image file.
- **Automatic Recipe Display:** Any recipe submitted by a user gets saved to the MySQL database and immediately appears on the recipes page.
- **Contact Us Form:** A page where visitors can send questions or feedback, which gets stored in the database.

1.4 Technology Stack

The project was developed using the required technologies for ICT 1209:

Layer	Technology	Description
Frontend Structure	HTML5 and Bootstrap 5	Responsive grid layout, navbar, cards, forms and carousel.
Styling	CSS3	Custom styling, card hover animations and sticky footer.
Client-Side Scripting	JavaScript (Vanilla)	Category filtering buttons, real-time live search and form validation.
Backend Scripting	PHP 8 (PDO)	Session handling, secure user login, form processing and image upload handling.
Database	MySQL (XAMPP)	Relational database storing users, recipes and contact messages.

2. System Design

2.1 Database Schema and Tables

The application uses a MySQL database named ‘recipe_book’. It contains three related tables to handle users, their submitted recipes and contact inquiries:

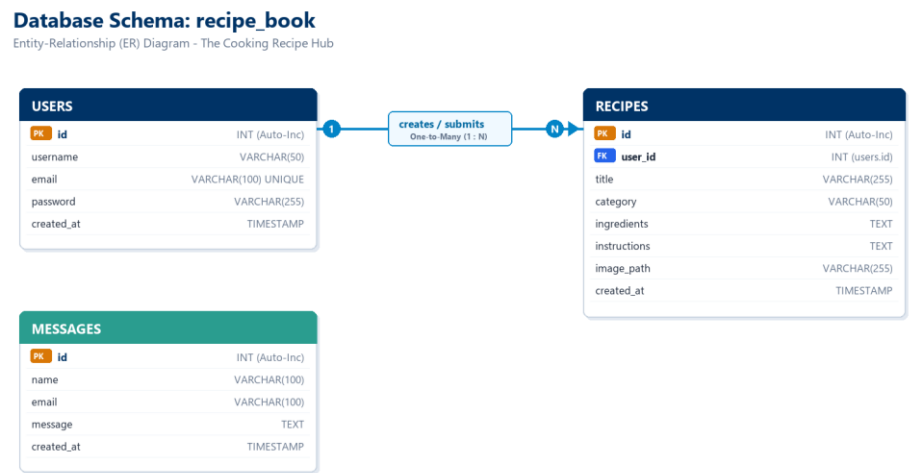


Figure 1: Entity-Relationship Diagram (ERD) of recipe_book Database

Database Relationship: One user can submit multiple recipes (1 : N). The ‘user_id’ field in the ‘recipes’ table connects each recipe to the user who uploaded it.

2.2 Database Table Specifications

Table	Field Name	Data Type	Key / Null	Description
users	id	INT	PK (Auto-Inc)	Unique User ID
users	username	VARCHAR(50)	NOT NULL	User display name
users	email	VARCHAR(100)	UNIQUE	User login email
users	password	VARCHAR(255)	NOT NULL	BCRYPT hashed password
users	created_at	TIMESTAMP	CURRENT_TIMESTAMP	Registration timestamp
recipes	id	INT	PK (Auto-Inc)	Unique Recipe ID
recipes	user_id	INT	FK (users.id)	ID of recipe author
recipes	title	VARCHAR(255)	NOT NULL	Title of the recipe
recipes	category	VARCHAR(50)	NOT NULL	Meal category (lunch, etc.)
recipes	ingredients	TEXT	NOT NULL	Required ingredients list
recipes	instructions	TEXT	NOT NULL	Preparation steps
recipes	image_path	VARCHAR(255)	NULL	Uploaded photo path
recipes	created_at	TIMESTAMP	CURRENT_TIMESTAMP	Submission timestamp
messages	id	INT	PK (Auto-Inc)	Unique Message ID
messages	name	VARCHAR(100)	NOT NULL	Sender full name

2.3 Folder Structure

Our project folder is organized cleanly as required by the course guidelines:

```
ict1209-mini-project/
├── auth/                # Login, Register and Logout PHP scripts
│   ├── login.php
│   ├── logout.php
│   └── register.php
├── css/                 # Custom stylesheet (style.css)
├── images/              # Logo and recipe images folder
│   ├── logo.png
│   └── Recipe images/
├── includes/            # Core backend PHP includes
│   ├── db.php           # PDO database connection
│   └── functions.php    # Session, sanitize and auth helper functions
├── js/                  # JavaScript files (filter.js, validation.js)
├── about.php            # About Us page
├── contact.php          # Contact form with database insertion
├── dashboard.php        # User recipe upload form
├── database.sql         # Database schema export file
├── index.php            # Home page with food carousel
├── README.md            # Project description and setup guide
└── recipes.php          # Recipe catalog with dynamic database cards
```

3. Implementation

This section describes the key features of the application along with screenshots showing the user interface.

3.1 Home Page and Interactive Carousel (index.php)

The home page introduces visitors to The Cooking Recipe Hub. It features an interactive Bootstrap 5 carousel showcasing traditional Sri Lankan dishes like Kiribath, Kottu and String Hoppers with automated cycling every 5 seconds. The navigation bar includes the brand logo and links to all main pages.

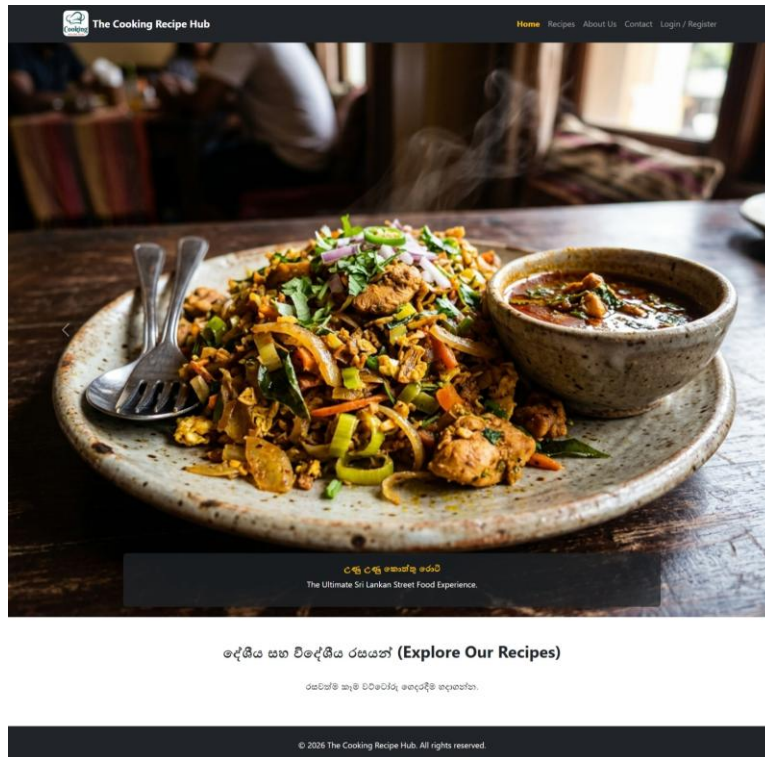


Figure 2: Home Page and Food Carousel

3.2 Recipe Catalog with Filtering and Search (recipes.php)

'recipes.php' displays all available recipes in a 3-column card layout. Users can filter recipes by clicking category buttons (All, Breakfast, Brunch, Lunch, Dinner, Snack and Dessert). A live search input lets users type recipe names to instantly find matching dishes using client-side JavaScript (js/filter.js).

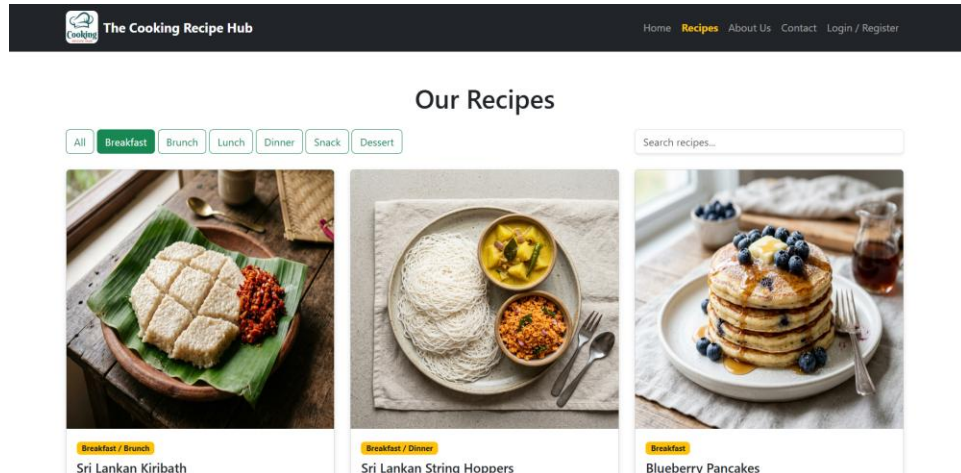


Figure 3: Recipe Catalog with Category Filter Buttons

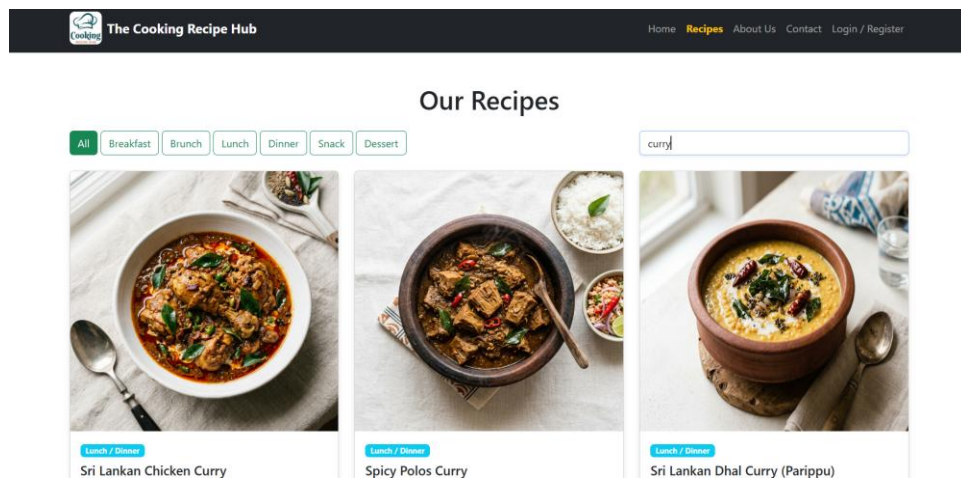
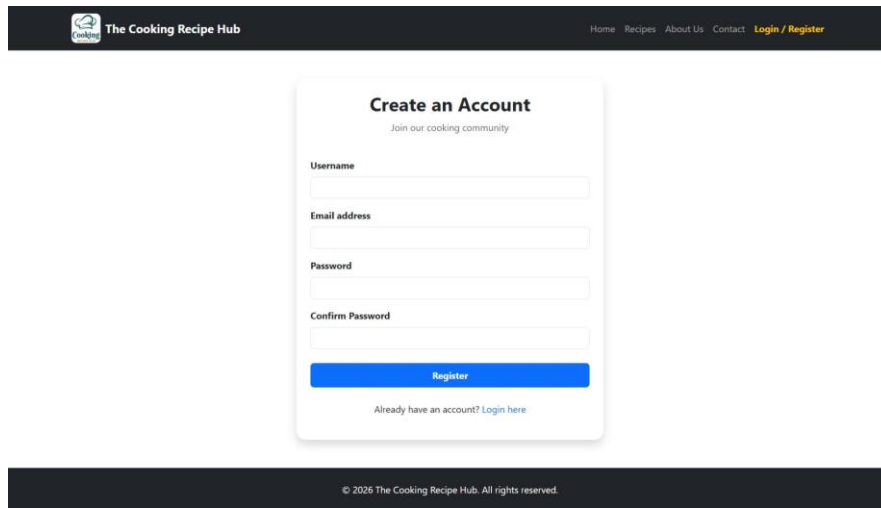


Figure 4: Real-Time Recipe Search Bar

3.3 User Authentication (register.php and login.php)

User registration and login are handled securely using PHP and MySQL. Passwords are encrypted using 'password_hash()' with BCrypt before saving to the database. When logging in, credentials are verified using 'password_verify()'. After successful login, 'session_regenerate_id()' is called for session security and the user is redirected to their Dashboard. The navigation bar dynamically updates to show "Dashboard" and "Logout (username)".



The image shows the 'Create an Account' registration form on 'The Cooking Recipe Hub' website. The form is centered on a white background with a light gray border. It includes input fields for Username, Email address, Password, and Confirm Password. A blue 'Register' button is at the bottom of the form. Below the button, there is a link: 'Already have an account? [Login here](#)'. The website's header is visible at the top, showing the logo and navigation links: Home, Recipes, About Us, Contact, and Login / Register. The footer at the bottom states: '© 2026 The Cooking Recipe Hub. All rights reserved.'

Figure 5: User Registration Page

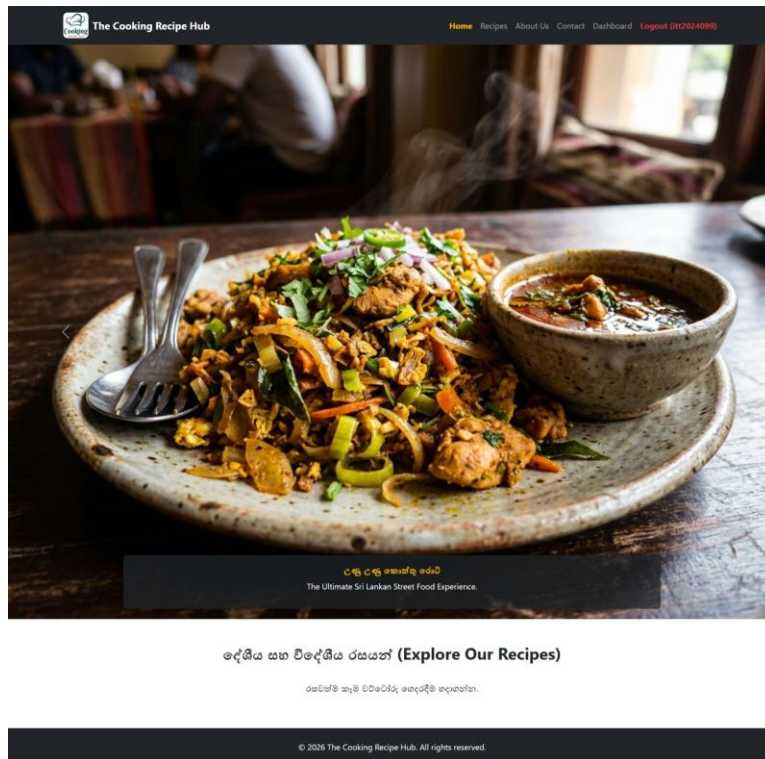
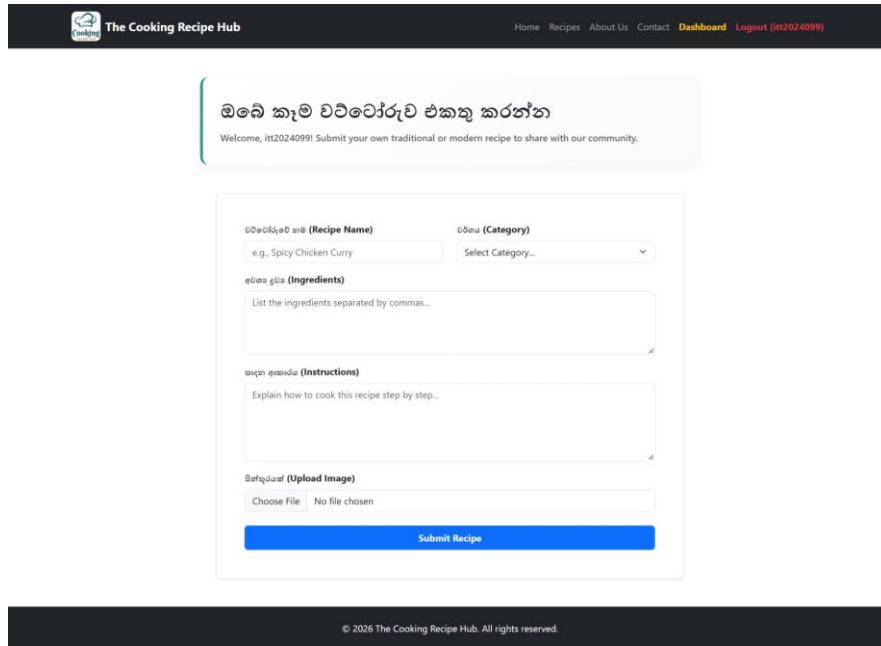


Figure 6: User Login Page and Dynamic Navbar

3.4 User Recipe Submission Dashboard (dashboard.php)

Only logged-in users can access ‘dashboard.php’. If an unauthenticated guest tries to visit this page, they are automatically redirected to ‘auth/login.php’. The dashboard contains a clean form where users enter the recipe title, select a category, list ingredients, write instructions and choose an image file. The backend checks file size (max 5MB) and type (JPG/PNG), renames the file uniquely with a timestamp, saves the photo to ‘images/Recipe images/’ and inserts the recipe into MySQL.



The screenshot shows the 'The Cooking Recipe Hub' dashboard. At the top, there's a navigation bar with links: Home, Recipes, About Us, Contact, Dashboard, and Logout (it2024099). Below the navigation bar, a welcome message reads: 'Welcome, it2024099! Submit your own traditional or modern recipe to share with our community.' The main form is titled 'ඔබේ කෑම විවිධත්වය එකතු කරන්න' (Add your own variety of food). It contains several input fields: 'විවිධත්වයේ නම (Recipe Name)' with a placeholder 'e.g., Spicy Chicken Curry', 'වර්ගය (Category)' with a dropdown menu 'Select Category...', 'අවශ්‍ය ද්‍රව්‍ය (Ingredients)' with a placeholder 'List the ingredients separated by commas...', 'යාලය ලියන්න (Instructions)' with a placeholder 'Explain how to cook this recipe step by step...', and 'ඡායාරූපය (Upload Image)' with a 'Choose File' button and 'No file chosen' text. A blue 'Submit Recipe' button is at the bottom of the form. The footer of the page reads '© 2026 The Cooking Recipe Hub. All rights reserved.'

Figure 7: User Dashboard and Recipe Submission Form

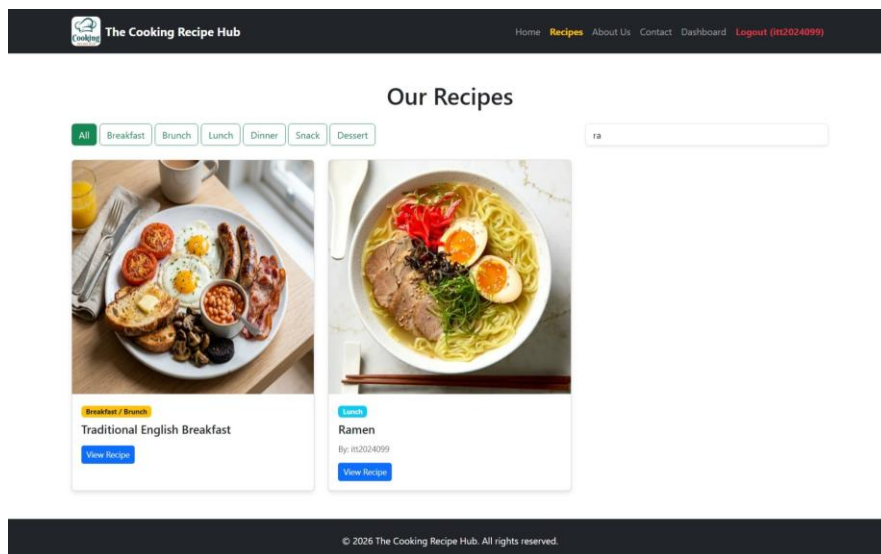


Figure 8: Dynamically Displayed User Recipe in Catalog

3.5 Contact Us Form (contact.php)

'contact.php' allows visitors to send feedback or inquiries. The form collects Name, Email and Message, validates the fields and stores the message in the 'messages' table. A green confirmation banner notifies the user of success.

The screenshot shows a web application header for 'The Cooking Recipe Hub' with navigation links: Home, Recipes, About Us, Contact, Dashboard, and Logout (rt2024099). The main content area is titled 'අපිට කියන්න (Contact Us)'. A green banner at the top of the form area says 'Thank you! Your message has been saved to the database successfully.' Below this is a form with three input fields: 'මම මගේ (Full Name)', 'විද්‍යාල ලිපිනය (Email)', and 'විස්තරය (Message)'. A green 'Send Message' button is at the bottom of the form.

Figure 9: Contact Us Form with Success Alert

3.6 Database Verification in phpMyAdmin

All user accounts, recipes and inquiries are stored and managed in the MySQL 'recipe_book' database via phpMyAdmin.

The screenshot shows the phpMyAdmin interface for the 'recipe_book' database. The top navigation bar includes links for Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines, Events, Triggers, and Designer. Below this is a 'Filters' section with a search box. The main area displays a table of database tables:

Table	Action	Rows	Type	Collation	Size	Overhead
messages	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 K B	-
recipes	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	32.0 K B	-
users	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	48.0 K B	-
3 tables	Sum	3	InnoDB	utf8mb4_general_ci	96.0 K B	0 B

Below the table list, there is a 'Create new table' form with fields for 'Table name' and 'Number of columns' (set to 4), and a 'Create' button.

Figure 10: phpMyAdmin Database Schema and Tables

4. Challenges and Solutions

During the development of the project, we encountered several practical challenges and solved them as follows:

4.1 Recipe Cards Vertical Stacking Issue

- **Problem:** When updating image paths in 'recipes.php', the recipe cards stopped displaying in a 3-column grid and stacked vertically in a single line.
- **Cause:** An opening '<div class="card-body">' tag had accidentally been deleted in several cards. This caused the Bootstrap flex column to close too early.
- **Solution:** We inspected the HTML, restored the missing '<div class="card-body">' tags and the 3-column grid worked properly again.

4.2 Database Connection Error on Recipes Page

- **Problem:** Newly added recipes were not showing on 'recipes.php' and clicking the category buttons stopped working.
- **Cause:** We used '\$pdo->query()' in 'recipes.php' without including 'includes/db.php' at the top. This caused a PHP fatal error which stopped the page halfway, preventing the JavaScript file ('filter.js') at the bottom from loading.
- **Solution:** We added 'require_once 'includes/db.php';' at the top of 'recipes.php'. The error was resolved, recipes loaded from the database and the filter buttons worked smoothly.

4.3 Local Server "404 Not Found" Error

- **Problem:** Opening 'http://localhost/ict1209-mini-project/index.php' in the browser gave an Apache 404 "Not Found" error.
- **Cause:** XAMPP's Apache web server only reads files inside 'C:\xampp\htdocs', but our project was saved in our personal documents folder.
- **Solution:** Instead of moving our project away from our Git folder, we opened Command Prompt as Administrator and created a directory junction: 'mklink /j "C:\xampp\htdocs\ict1209-mini-project" "C:\path\to\project"'. This let Apache access our project directly.

4.4 Image Upload and Filename Duplication

- **Problem:** When multiple users upload recipe images with the same common name (such as curry.jpg) files could overwrite each other.
- **Solution:** We validated uploaded files with 'getimagesize()' to make sure they are genuine images, restricted file sizes to 5MB and added a timestamp prefix ('time() . '_' . filename') so every saved image has a unique name.

4.5 Dashboard Link Highlight in Navigation Bar

- **Problem:** The Dashboard link in the navbar was always highlighted in yellow ('text-warning'), even when the user was on the Home or Contact page.
- **Solution:** We removed the yellow highlight class from 'index.php', 'recipes.php', 'about.php' and 'contact.php' and kept it active only inside 'dashboard.php'.

5. Individual Contribution

Both group members contributed equally across all three phases of the project as summarized below:

Project Phase / Task	L.P.D.T Senarathna (ITT/2024/099)	R.M.D.S. Kumara (ITT/2024/056)
Phase 1: Proposal and Wireframes	• Created wireframes for Home, Catalog and Dashboard. • Outlined planned user features.	• Wrote the initial Project Proposal document. • Researched the required technology stack.
Phase 2: Frontend Layout and Design	• Developed the Recipe Catalog ('recipes.php'). • Implemented JavaScript category filtering. • Styled recipe cards and fixed grid layout.	• Implemented the food carousel ('index.php'). • Designed Contact Us frontend layout. • Structured base CSS styles and sticky footer.
Phase 3: PHP and MySQL Backend	• Built the User Dashboard ('dashboard.php'). • Handled image uploads and file validation. • Displayed database recipes on catalog page. • Updated dynamic navbar session state.	• Designed the MySQL database ('database.sql'). • Configured PDO connection ('includes/db.php'). • Built User Registration and Login with BCrypt. • Built Contact form database insertion.
Documentation and Testing	• Git commits, branch merging and repo sync. • Created Apache webroot directory junction. • Documented challenges and setup steps.	• Maintained README.md and inline code comments. • Prepared SQL export file and test data. • Documented system design and schema.

5.1 Summary of Individual Tasks

L.P.D.T Senarathna (ITT/2024/099): Senarathna focused on the Recipe Submission Dashboard ('dashboard.php'), image file uploading logic, dynamic recipe display in the catalog, fixing grid layout bugs and managing dynamic navigation states.

R.M.D.S. Kumara (ITT/2024/056): Kumara focused on designing the MySQL database schema ('database.sql'), establishing the PDO database connection ('db.php'), building User Registration and Login with secure password hashing ('register.php', 'login.php' and 'logout.php') and saving contact inquiries ('contact.php').

6. Setup Instructions (Running via XAMPP)

To set up and run “The Cooking Recipe Hub” locally on a computer using XAMPP, follow these simple steps:

- **Prerequisites:** Download and install XAMPP (with Apache and MySQL) from <https://www.apachefriends.org/>.
- **Step 1: Put Project in XAMPP htdocs** - Copy the project folder ‘ict1209-mini-project’ into ‘C:\xampp\htdocs\’.
(Alternatively, if the folder is in Documents, open Command Prompt as Administrator and run: `mklink /j “C:\xampp\htdocs\ict1209-mini-project” “C:\path\to\your\folder”`)
- **Step 2: Start Apache and MySQL** - Open XAMPP Control Panel and click the ‘Start’ button next to both Apache and MySQL. Make sure both turn green.
- **Step 3: Import the Database** - Open your web browser and go to <http://localhost/phpmyadmin/>.
 - Click on the ‘Import’ tab at the top.
 - Click ‘Choose File’ and select ‘database.sql’ from the project folder.
 - Click ‘Go’ at the bottom. This will automatically create the ‘recipe_book’ database and its 3 tables.
- **Step 4: Check Database Settings** - Check ‘includes/db.php’ in the project. The default settings match standard XAMPP (host: localhost, dbname: recipe_book, user: root, password: empty).
- **Step 5: Open and Test the Website** - Open your browser and visit: <http://localhost/ict1209-mini-project/index.php>
You can now explore recipes, register a new account, log in, submit a recipe via the Dashboard and send a message through the Contact page.

7. References

- W3Schools, “PHP 8 and MySQL Database Tutorial,” W3Schools Online Web Tutorials, 2026. [Online]. Available: <https://www.w3schools.com/php/>.
- Bootstrap, “Bootstrap v5.3 Documentation: Grid, Components and Utilities,” 2026. [Online]. Available: <https://getbootstrap.com/docs/5.3/>.
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